

Review: "Evolution of Reinforcement Learning Agents Using the Genetic Algorithm"

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1. Introduction

The paper applies a genetic algorithm to reinforcement learning, evolving the parameters of a neural policy instead of training them by gradient descent. The motivation is stated plainly: in reinforcement learning the reward arrives rarely and late, so the gradient carries little information and an agent can wander for a long time before learning anything. The authors ask whether a gradient-free search does better, and test it on the CartPole-v0 control task.

2. Methodology

The agent is a two-layer network in PyTorch that takes the four state variables of CartPole and produces two actions; its weights and biases form the genome. A population of 500 agents is initialised at random, each is evaluated by its mean reward over three episodes, and the twenty best are kept as parents. Each new generation copies those parents at random and adds Gaussian noise scaled by 0.02, with the single best agent carried over unchanged. The loop runs for 200 generations.

3. Results

The mean reward across the population rises from 22 in the first generation to 197 in the two hundredth. The first agent to solve the task appears at generation 55, twenty agents reach the maximum by generation 71, and from generation 139 onward the population average stays above 195. In a sample of a hundred agents, 98 obtain the maximum reward of 200.

4. Key Insights

Insight 1. The case for a gradient-free method rests on a property of the problem rather than of the optimiser. I measured the same trade-off in my own work: on a smooth convex loss a genetic algorithm reached the same optimum as Adam but needed tens of thousands of evaluations against a thousand gradient steps. Evolution earns its cost exactly where this paper places it — where the gradient is sparse, noisy or absent.

Insight 2. The procedure that runs is simpler than the one described. Crossover is presented as a core operator but is never applied; what the experiment performs is truncation selection with Gaussian perturbation and one unperturbed elite. That elite copy keeps the best fitness monotone across generations, and it is the component I would not omit.

Insight 3. The noise scale is cast explicitly as the counterpart of a learning rate, tuned by hand to 0.02, and the authors admit there is no principled way to choose it. This matches what I found tuning a mutation rate myself: a narrow band converges reliably, everything outside it degrades sharply.

5. Conclusion

The contribution is a clearly documented, reproducible demonstration that a simple evolutionary loop can solve a control task without computing a single gradient. Two gaps limit it. The novelty the paper claims — initialising the population with domain knowledge — is described at length but not used, since the experiments initialise at random. And although the stated motivation is that gradient methods perform poorly here, no gradient baseline is run, so the comparison the argument depends on is never measured.